

COSC/MATH 4340: Numerical Methods for Differential Equations

Partial Differential Equations (PDEs)

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Introduction

Thus far we have considered *ordinary differential equations (ODEs)* a variable $y(t)$ changes in time t or systems of ODEs in which a set of variables $y_i(t)$ change in time:

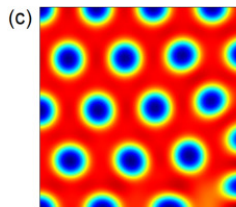
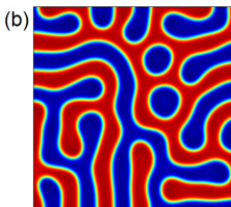
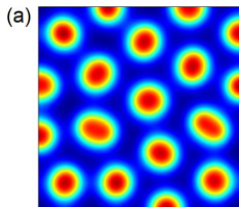
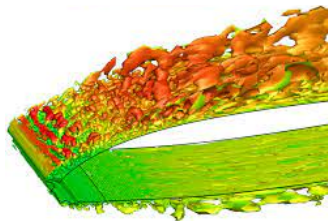
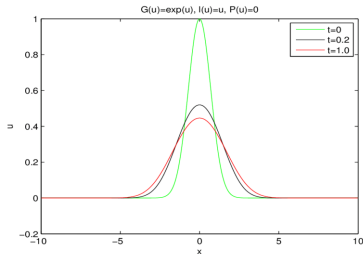
$$\frac{d}{dt}\mathbf{y} = f(t, \mathbf{y})$$

with initial condition $\mathbf{y}(t_0) = \mathbf{y}_0$ which has a solution $\mathbf{y}(t)$ over some range $t \in [t_0, T]$.

We now study *partial differential equations (PDEs)* in which a function $u(x, t)$ depends on two or more independent and continuous variables. Here, $x \in [a, b]$ often represents a spatial dimension and $t \in [0, T]$ represents time.



Introduction





Introduction

Definition

A **2nd-order partial differential equation (PDEs)** takes the form

$$Au_{xx} + Bu_{xy} + xu_{yy} + F(u_x, u_y, u, x, y) = 0,$$

where subscripts indicate partial derivatives, i.e.,

$$u_{xy} = \frac{\partial}{\partial x} \frac{\partial}{\partial y} u(x, y).$$



Types of PDEs

Given a PDE

$$Au_{xx} + Bu_{xy} + Cu_{yy} + F(u_x, u_y, u, x, y) = 0,$$

we can we can classify types of dynamics as follows:

- ▶ **Parabolic PDE** if $B^2 - 4AC = 0$: diffusion-like dynamics that dissipate with time
- ▶ **Hyperbolic PDE** if $B^2 - 4AC > 0$: wave-like dynamics that potentially oscillate
- ▶ **Elliptic PDE** if $B^2 - 4AC < 0$: curved solutions that fit boundary conditions

The Heat Equation

Definition

The heat equation is given by $u_t = Du_{xx}$, or equivalently

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2},$$

where $u(x, t)$ denotes the temperature at location x at time t and $D > 0$ is the **diffusion coefficient**

Note that the heat equation is a parabolic equation, since $B^2 - 4AC = 0$ (i.e., $B = C = 0$ and $A = D$).



The Heat Equation

The full heat equation must be specified with an initial condition and boundary conditions:

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}$$

$$u(x, 0) = f(x), \quad x \in [a, b] \quad \text{initial temperature profile}$$

$$u(a, t) = l(t), \quad t \in [t_0, T] \quad \text{temp on the left boundary}$$

$$u(b, t) = r(t), \quad t \in [t_0, T] \quad \text{temp on the right boundary}$$



(Forward) Finite Difference Method

We discretize both the space and time domains with meshes

$$t_j = t_0 + jk, \quad j = 0, 1, \dots, N$$

$$x_i = a + ih, \quad i = 0, 1, \dots, n,$$

so that there are $N + 1$ time points with step size $k = (T - t_0)/N$ and $n + 1$ spatial points with step size $h = (b - a)/n$.

Next, we let $u(x_i, t_j)$ denote the true function at position x_i at time t_j , and we let $u_{i,j}$ denote our approximation.



(Forward) Finite Difference Method

Considering the PDE $\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}$, we approximate both sides using finite differences:

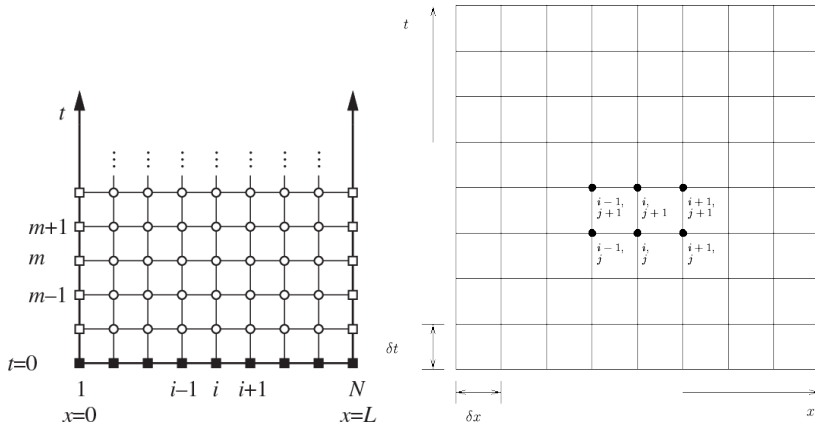
$$\begin{aligned}\frac{\partial u}{\partial t} &\approx \frac{u(x_i, t_j + k) - u(x_i, t_j)}{k} + \mathcal{O}(k) \approx \frac{u_{i,j+1} - u_{i,j}}{k} \\ \frac{\partial^2 u}{\partial x^2} &\approx \frac{u(x_i + h, t_j) - 2u(x_i, t_j) + u(x_i - h, t_j)}{h^2} + \mathcal{O}(h^2) \\ &\approx \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2}\end{aligned}$$

Combining these yields an “explicit” finite difference equation

$$u_{i,j+1} = \sigma u_{i+1,j} + (1 - 2\sigma)u_{i,j} + \sigma u_{i-1,j},$$

where $\sigma = Dk/h^2$. Note the approximation is 1st order in k and 2nd order in h .

(Forward) Finite Difference Method



(See notes and code for an example.)



(Forward) Finite Difference Method

Letting $\mathbf{u}_j = [u_{1,j}, u_{2,j}, \dots, u_{n-1,j}]^T$ and $\mathbf{s}_j = [u_{0,j}, 0, \dots, 0, u_{n,j}]^T$, the forward finite difference equation takes the form

$$\mathbf{u}_{j+1} = A\mathbf{u}_j + \sigma\mathbf{s}_j,$$

where

$$A = \begin{bmatrix} 1-2\sigma & \sigma & 0 & \dots & 0 \\ \sigma & 1-2\sigma & \sigma & \ddots & \vdots \\ 0 & \sigma & 1-2\sigma & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \sigma \\ 0 & \dots & 0 & \sigma & 1-2\sigma \end{bmatrix}.$$



Von Neumann Stability

Definition

The iterative equation $\mathbf{u}_{j+1} = A\mathbf{u}_j + \sigma\mathbf{s}_j$ is **Von Neumann stable** if the spectral radius is less than one: $\rho(A) < 1$.

Theorem

The Forward Finite Difference Method is stable if $\sigma < 1/2$.

Definition

A method is called **conditionally stable** if its stability depends on the step sizes h and k .

(See notes and code.)



(Backward) Finite Difference Method

Letting $\mathbf{u}_j = [u_{1,j}, u_{2,j}, \dots, u_{n-1,j}]^T$ and $\hat{\mathbf{s}}_{j+1} = [u_{0,j+1}, 0, \dots, 0, u_{n,j+1}]^T$, the backward finite difference equation takes the form

$$\hat{A}\mathbf{u}_{j+1} = \mathbf{u}_j + \sigma\hat{\mathbf{s}}_{j+1},$$

where

$$\hat{A} = \begin{bmatrix} 1 + 2\sigma & -\sigma & 0 & \dots & 0 \\ -\sigma & 1 + 2\sigma & -\sigma & \ddots & \vdots \\ 0 & -\sigma & 1 + 2\sigma & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \sigma \\ 0 & \dots & 0 & -\sigma & 1 + 2\sigma \end{bmatrix}.$$

(See notes.)



Von Neumann Stability

Theorem

The Von Neumann stability of the Backward Finite Difference Method is equivalent to the stability for the iteration $\mathbf{u}_{j+1} = \hat{A}^{-1}\mathbf{u}_j + \sigma\hat{A}^{-1}\mathbf{s}_{j+1}$, requiring $\rho(\hat{A}^{-1}) < 1$, which is guaranteed for any σ .

Definition

A method is called **unconditionally stable** if its stability does not depend on the step sizes h and k .

(See notes.)



Types of Boundary Conditions

Definition

The heat equation is said to have **Dirichlet boundary conditions** if one specifies the values at a and b : $u(a, t) = l(t)$ and $u(b, t) = r(t)$.

Definition

The heat equation is said to have **Neumann boundary conditions** if one specifies the derivatives at a and b : $u_x(a, t) = l(t)$ and $u_x(b, t) = r(t)$.

Definition

In general, setting the boundary conditions to zero is called **homogeneous boundary conditions**. For example, setting $u(a, t) = u(b, t) = 0$ or $u_x(a, t) = u_x(b, t) = 0$.



Crank-Nicolson Method

The Crank-Nicolson Method combines the forward and backward Finite Difference methods

$$\frac{\partial u}{\partial t} \approx \frac{u_{i,j+1} - u_{i,j}}{k}$$
$$\frac{\partial^2 u}{\partial x^2} \approx \frac{1}{2} \left[\frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2} + \frac{u_{i+1,j+1} - 2u_{i,j+1} + u_{i-1,j+1}}{h^2} \right]$$

and takes the matrix form

$$(I + \hat{A})\mathbf{u}_{j+1} = (I + A)\mathbf{u}_j + \sigma(\hat{\mathbf{s}}_{j+1} + \mathbf{s}_j),$$

(See notes.)



Crank-Nicolson Method

Theorem

The Von Neumann stability of the Crank-Nicolson Method is equivalent to the stability for the iteration

$\mathbf{u}_{j+1} = (I + \hat{A})^{-1}(I + A)\mathbf{u}_j + \sigma(I + \hat{A})^{-1}(\mathbf{s}_{j+1} + \mathbf{s}_j)$, requiring $\rho\left((I + \hat{A})^{-1}(I + A)\right) < 1$, which is guaranteed for any σ .

Theorem

The Crank-Nicolson Method is unconditionally stable

(See notes.)



Accuracy of Finite Difference Methods

- ▶ The Forward Finite Difference Method and Backward Finite Difference Method both have local truncation errors that are $\mathcal{O}(h^2) + \mathcal{O}(k)$.
- ▶ The Crank-Nicolson Method has local truncation errors that are $\mathcal{O}(h^2) + \mathcal{O}(k^2)$.

(See textbook pages 388-389 for derivation.)